```sql
-- 34 IMPORT COUNT COMPARED WITH ORIGINAL DATASET REFERENCE COUNTS
-- Reference: customers 99,441; orders 99,441; payments 103,886.
SELECT 'customers' table_name,COUNT(*) imported_rows,99441 reference_rows,COUNT(*)-99441 difference FROM customers
UNION ALL SELECT 'orders',COUNT(*),99441,COUNT(*)-99441 FROM orders
UNION ALL SELECT 'order_payments',COUNT(*),103886,COUNT(*)-103886 FROM order_payments;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

table_name	imported_rows	reference_rows	difference
customers	99441	99441	0
orders	99441	99441	0
order_payments	103886	103886	0

```sql
-- 33 ORDERS WITH MULTIPLE PAYMENT ROWS (can be legitimate)
SELECT order_id,COUNT(*) payment_rows,SUM(payment_value) total_payment_value
FROM order_payments GROUP BY order_id HAVING COUNT(*)>1
ORDER BY payment_rows DESC LIMIT 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

order_id	payment_rows	total_payment_value
fa65dad1b0e818e3ccc5cb0e39231352	29	457.99
ccf804e764ed5650cd8759557269dc13	26	62.680000000000014
285c2e15bebd4ac83635ccc563dc71f4	22	40.85
895ab968e7bb0d5659d16cd74cd1650c	21	161.32000000000002
ee9ca989fc93ba09a6eddc250ce01742	19	82.73000000000002
fedcd9f7ccdc8cba3a18defedd1a5547	19	205.73999999999998
21577126c19bf11a0b91592e5844ba78	15	86.99000000000001
4bfcba9e084f46c8e3cb49b0fa6e6159	15	740.76
3c58bffb70dcf45f12bdf66a3c215905	14	100.57000000000001
4689b1816de42507a7d63a4617383c59	14	529.5500000000001
4fb76fa13b108a0d047848342 1b0992c	13	107.27

```sql
-- 32 ORDERS WITH NO PAYMENT ROW (review statuses may not require payment)
SELECT o.order_id,o.order_status FROM orders o
LEFT JOIN order_payments p ON o.order_id=p.order_id
WHERE p.order_id IS NULL LIMIT 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	order_status
bfbd0f9bdef84302105ad712db648a6c	delivered

```sql
-- 31 ORDER STATUS COUNTS AND PERCENTAGES
SELECT order_status,COUNT(*) order_count,
ROUND(100.0*COUNT(*)/(SELECT COUNT(*) FROM orders),2) pct_of_orders
FROM orders GROUP BY order_status ORDER BY order_count DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_status	order_count	pct_of_orders
delivered	96478	97.02
shipped	1107	1.11
canceled	625	0.63
unavailable	609	0.61
invoiced	314	0.32
processing	301	0.30
created	5	0.01
approved	2	0.00

```sql
-- 30 TOP 20 HIGHEST PAYMENTS (potential outliers; review)
SELECT order_id,payment_sequential,payment_type,payment_value
FROM order_payments ORDER BY payment_value DESC LIMIT 20;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

order_id	payment_sequential	payment_type	payment_value
03caa2c082116e1d31e67e9ae3700499	1	credit_card	13664.08
736e1922ae60d0d6a89247b851902527	1	boleto	7274.88
0812eb902a67711a1cb742b3cdaa65ae	1	credit_card	6929.31
fefacc66af859508bf1a7934eab1e97f	1	boleto	6922.21
f5136e38d1a14a4dbd87dff67da82701	1	boleto	6726.66
2cc9089445046817a7539d90805e6e5a	1	boleto	6081.54
a96610ab360d42a2e5335a3998b4718a	1	credit_card	4950.34
b4c4b76c642808cbe472a32b86cddc95	1	credit_card	4809.44
199af31afc78c699f0dbf71fb178d4d4	1	credit_card	4764.34
8dbc85d1447242f3b127dda390d56e19	1	credit_card	4681.78
426a9742b533fc6fed17d1fd6d143d7e	1	credit_card	4513.32

```sql
-- 29 PAYMENT VALUE DISTRIBUTION SUMMARY
SELECT MIN(payment_value) min_value,MAX(payment_value) max_value,
AVG(payment_value) avg_value,STDDEV_POP(payment_value) population_stddev,
COUNT(*) payment_rows FROM order_payments;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

min_value	max_value	avg_value	population_stddev	payment_rows
0	13664.08	154.10038041698365	217.49301707022187	103886

```sql
-- 28 ZIP PREFIX OUT OF 0..99999 RANGE
SELECT customer_id,customer_zip_code_prefix FROM customers
WHERE customer_zip_code_prefix IS NULL OR customer_zip_code_prefix<0
OR customer_zip_code_prefix>99999 LIMIT 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	customer_zip_code_prefix

```sql
-- 27 MISSING/BLANK CUSTOMER CITY
SELECT customer_id,customer_city FROM customers
WHERE customer_city IS NULL OR TRIM(customer_city)='' LIMIT 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	customer_city

```sql
-- 26 UNEXPECTED BRAZILIAN STATE CODES
SELECT customer_id,customer_state FROM customers
WHERE customer_state IS NULL OR UPPER(TRIM(customer_state)) NOT IN
('AC','AL','AP','AM','BA','CE','DF','ES','GO','MA','MT','MS','MG','PA','PB','PR','PE','PI','RJ','RN','RS','RO','RR','SC','SP','SE','TO')
LIMIT 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	customer_state

```sql
-- 25 CUSTOMER STATE PROFILE
SELECT customer_state,COUNT(*) records FROM customers
GROUP BY customer_state ORDER BY records DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_state	records
SP	41746
RJ	12852
MG	11635
RS	5466
PR	5045
SC	3637
BA	3380
DF	2140
ES	2033
GO	2020
PE	1652

```sql
-- 24 LATE DELIVERIES VS ESTIMATED DATE
SELECT order_id,order_status,order_delivered_customer_date,order_estimated_delivery_date
FROM orders WHERE order_delivered_customer_date IS NOT NULL
AND order_estimated_delivery_date IS NOT NULL
AND order_delivered_customer_date>order_estimated_delivery_date LIMIT 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

order_id	order_status	order_delivered_customer_date	order_estimated_delivery_date
203096f03d82e0dffbc41ebc2e2bcfb7	delivered	2017-10-09 22:23:46	2017-09-28 00:00:00
fbf9ac61453ac646ce8ad9783d7d0af6	delivered	2018-03-21 22:03:54	2018-03-12 00:00:00
8563039e855156e48fccee4d611a3196	delivered	2018-03-20 00:59:25	2018-03-20 00:00:00
6ea2f835b4556291ffdc53fa0b3b95e8	delivered	2017-12-28 18:59:23	2017-12-21 00:00:00
66e4624ae69e7dc89bd50222b59f581f	delivered	2018-04-03 13:28:46	2018-04-02 00:00:00
a685d016c8a26f71a0bb67821070e398	delivered	2017-04-06 13:37:16	2017-03-30 00:00:00
6a0a8bfbbe700284feb0845d95e0867f	delivered	2017-12-28 19:43:00	2017-12-11 00:00:00
a5474c0071dd5d1074e12d417078bbd0	delivered	2018-08-03 19:28:47	2018-08-02 00:00:00
9d531c565e28c3e0d756192f84d8731f	delivered	2018-01-23 21:38:52	2017-12-22 00:00:00
8fc207e94fa91a7649c5a5dab690272a	delivered	2018-01-20 13:42:22	2017-12-19 00:00:00
33a3edb84b9df4cb495468 59b990ac6d	delivered	2018-03-22 00:03:53	2018-03-16 00:00:00

```sql
-- 23 APPROVAL BEFORE PURCHASE (review)
SELECT order_id,order_purchase_timestamp,order_approved_at FROM orders
WHERE order_approved_at IS NOT NULL AND order_approved_at<order_purchase_timestamp LIMIT 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

order_id	order_purchase_timestamp	order_approved_at
00b1cb0320190ca0daa2c88b35206009	2018-08-28 15:26:39	
ed3efbd3a87bea76c2812c66a0b32219	2018-09-20 13:54:16	
df8282afe61008dc26c6c31011474d02	2017-03-04 12:14:30	
8d4c637f1accf7a88a4555f02741e606	2018-08-29 16:27:49	
7a9d4c7f9b068337875b95465330f2fc	2017-05-01 16:12:39	
ddaec6fff982b13e7e048b627a11d6da	2016-10-04 19:41:32	
5290c34bd38a8a095b885f13958db1e1	2018-08-21 10:25:18	
03310aa823a66056268a3bab36e827fb	2018-08-07 16:33:59	
4c8b9947280829d0a8b7e81cc249b875	2018-08-09 14:54:47	
b13ea375fe9c728832688264638f84cf	2018-08-22 18:52:29	
ea844c92cf978ea23321fa7fe5871761	2018-09-13 09:56:12	

```sql
-- 22 DELIVERY DATE PRESENT WITH NON-DELIVERED STATUS (review)
SELECT order_id,order_status,order_delivered_customer_date FROM orders
WHERE order_delivered_customer_date IS NOT NULL
AND LOWER(TRIM(order_status))<>'delivered' LIMIT 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

order_id	order_status	order_delivered_customer_date
136cce7faa42fdb2cefd53fdc79a6098	invoiced	
ee64d42b8cf066f35eac1cf57de1aa85	shipped	
0760a852e4e9d89eb77bf631eaaf1c84	invoiced	
15bed8e2fec7fdbadb186b57c46c92f2	processing	
6942b8da583c2f9957e990d028607019	shipped	
36530871a5e80138db53bcfd8a104d90	shipped	
4d630f57194f5aba1a3d12ce23e71cd9	shipped	
8e24261a7e58791d10cb1bf9da94df5c	unavailable	
3b4ad687e7e5190db827e1ae5a8989dd	shipped	
b68d69564a79dea4776afa33d1d2fcab	shipped	

```sql
-- 21 DELIVERED STATUS WITH NULL CUSTOMER DELIVERY DATE
SELECT order_id,order_status,order_delivered_customer_date FROM orders
WHERE LOWER(TRIM(order_status))='delivered'
AND order_delivered_customer_date IS NULL LIMIT 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	order_status	order_delivered_customer_date

```sql
-- 20 CUSTOMER DELIVERY BEFORE CARRIER DATE
SELECT order_id,order_status,order_delivered_carrier_date,order_delivered_customer_date
FROM orders WHERE order_delivered_customer_date IS NOT NULL
AND order_delivered_carrier_date IS NOT NULL
AND order_delivered_customer_date<order_delivered_carrier_date LIMIT 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

order_id	order_status	order_delivered_carrier_date	order_delivered_customer_date
ee64d42b8cf066f35eac1cf57de1aa85	shipped	2018-06-05 14:32:00	
6942b8da583c2f9957e990d028607019	shipped	2018-01-11 19:39:23	
36530871a5e80138db53bcfd8a104d90	shipped	2017-05-11 13:21:47	
4d630f57194f5aba1a3d12ce23e71cd9	shipped	2017-11-22 17:28:34	
3b4ad687e7e5190db827e1ae5a8989dd	shipped	2018-07-04 15:20:00	
b68d69564a79dea4776afa33d1d2fcab	shipped	2018-03-05 16:10:13	
f373335aac9a659de916f7170b8bc07a	shipped	2018-03-20 21:08:28	
5869074071e0f56e9085dc3742c97b2a	shipped	2018-06-12 14:09:00	
6147a4d49752afd35a119d794e481588	shipped	2017-05-02 15:29:33	
95e42e6aaf6264cd3e77c06b32dc3003	shipped	2017-11-24 19:07:43	
10918fe4d665fda80034712895c2b92f	shipped	2017-07-04 11:27:21	

Result Grid
Form Editor
Field Types

```sql
-- 19 DELIVERY DATE BEFORE PURCHASE
SELECT order_id,order_status,order_purchase_timestamp,
order_delivered_carrier_date,order_delivered_customer_date
FROM orders WHERE
(order_delivered_carrier_date IS NOT NULL AND order_delivered_carrier_date<order_purchase_timestamp)
OR (order_delivered_customer_date IS NOT NULL AND order_delivered_customer_date<order_purchase_timestamp)
LIMIT 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

er_id	order_status	order_purchase_timestamp	order_delivered_carrier_date	order_delivered_customer_date
ce7faa42fdb2cefd53fdc79a6098	invoiced	2017-04-11 12:22:08		
d42b8cf066f35eac1cf57de1aa85	shipped	2018-06-04 16:44:48	2018-06-05 14:32:00	
a852e4e9d89eb77bf631eaaf1c84	invoiced	2018-08-03 17:44:42		
d8e2fec7fdbadb186b57c46c92f2	processing	2017-09-03 14:22:03		
b8da583c2f9957e990d028607019	shipped	2018-01-10 11:33:07	2018-01-11 19:39:23	
0871a5e80138db53bcfd8a104d90	shipped	2017-05-09 11:48:37	2017-05-11 13:21:47	
0f57194f5aba1a3d12ce23e71cd9	shipped	2017-11-17 19:53:21	2017-11-22 17:28:34	
261a7e58791d10cb1bf9da94df5c	unavailable	2017-11-16 15:09:28		
d687e7e5190db827e1ae5a8989dd	shipped	2018-06-28 12:52:15	2018-07-04 15:20:00	
69564a79dea4776afa33d1d2fcab	shipped	2018-02-28 08:57:03	2018-03-05 16:10:13	

Result Grid | Form Editor | Field Types

```sql
-- 18 INVALID INSTALLMENT COUNTS (zero/negative)
SELECT order_id,payment_sequential,payment_installments FROM order_payments
WHERE payment_installments<=0 ORDER BY payment_installments;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	payment_sequential	payment_installments
744bade1fcf9ff3f31d860ace076d422	2	0
1a57108394169c0b47d8f876acc9ba2d	2	0

```sql
-- 17 ZERO PAYMENT AMOUNTS (flag for review)
SELECT order_id,payment_sequential,payment_type,payment_value FROM order_payments
WHERE payment_value=0 LIMIT 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	payment_sequential	payment_type	payment_value
8bcbe01d44d147f901cd3192671144db	4	voucher	0
fa65dad1b0e818e3ccc5cb0e39231352	14	voucher	0
6ccb433e00daae1283ccc956189c82ae	4	voucher	0
4637ca194b6387e2d538dc89b124b0ee	1	not_defined	0
00b1cb0320190ca0daa2c88b35206009	1	not_defined	0
45ed6e85398a87c253db47c2d9f48216	3	voucher	0
fa65dad1b0e818e3ccc5cb0e39231352	13	voucher	0
c8c528189310eaa44a745b8d9d26908b	1	not_defined	0
b23878b3e8eb4d25a158f57d96331b18	4	voucher	0

```sql
-- 16 NEGATIVE PAYMENT AMOUNTS
SELECT order_id,payment_sequential,payment_value FROM order_payments
WHERE payment_value<0 ORDER BY payment_value;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	payment_sequential	payment_value

```sql
-- 15 INVALID/UNEXPECTED PAYMENT TYPE
SELECT order_id,payment_sequential,payment_type FROM order_payments
WHERE payment_type IS NULL OR LOWER(TRIM(payment_type)) NOT IN
('credit_card','boleto','voucher','debit_card','not_defined') LIMIT 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	payment_sequential	payment_type

```sql
-- 14 PAYMENT TYPE PROFILE
SELECT payment_type, COUNT(*) records FROM order_payments
GROUP BY payment_type ORDER BY records DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

payment_type	records
credit_card	76795
boleto	19784
voucher	5775
debit_card	1529
not_defined	3

```sql
-- 13 INVALID/UNEXPECTED ORDER STATUS
SELECT order_id,order_status FROM orders
WHERE order_status IS NULL OR LOWER(TRIM(order_status)) NOT IN
('delivered','shipped','canceled','unavailable','invoiced','processing','created','approved')
LIMIT 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	order_status

```sql
-- 12 ORDER STATUS PROFILE
SELECT order_status, COUNT(*) records FROM orders
GROUP BY order_status ORDER BY records DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_status	records
delivered	96478
shipped	1107
canceled	625
unavailable	609
invoiced	314
processing	301
created	5
approved	2

```sql
-- 11 ORPHAN PAYMENT ORDER REFERENCES: SAMPLE
SELECT p.order_id,p.payment_sequential,p.payment_type,p.payment_value
FROM order_payments p LEFT JOIN orders o ON p.order_id=o.order_id
WHERE p.order_id IS NULL OR o.order_id IS NULL LIMIT 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	payment_sequential	payment_type	payment_value

```sql
-- 10 ORPHAN PAYMENT ORDER REFERENCES: COUNT
SELECT COUNT(*) orphan_payment_order_rows FROM order_payments p
LEFT JOIN orders o ON p.order_id=o.order_id
WHERE p.order_id IS NULL OR o.order_id IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

orphan_payment_order_rows
0

```sql
-- 09 ORPHAN ORDER CUSTOMER REFERENCES: SAMPLE
SELECT o.order_id, o.customer_id FROM orders o
LEFT JOIN customers c ON o.customer_id=c.customer_id
WHERE o.customer_id IS NULL OR c.customer_id IS NULL LIMIT 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	customer_id

```sql
-- 08 ORPHAN ORDER CUSTOMER REFERENCES: COUNT
SELECT COUNT(*) orphan_order_customer_rows FROM orders o
LEFT JOIN customers c ON o.customer_id=c.customer_id
WHERE o.customer_id IS NULL OR c.customer_id IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

orphan_order_customer_rows
0

```sql
-- 07 DUPLICATE PAYMENT COMPOSITE KEY
SELECT order_id, payment_sequential, COUNT(*) occurrences FROM order_payments
GROUP BY order_id, payment_sequential HAVING COUNT(*)>1 ORDER BY occurrences DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	payment_sequential	occurrences

```sql
-- 06 DUPLICATE ORDER ID
SELECT order_id, COUNT(*) occurrences FROM orders
GROUP BY order_id HAVING COUNT(*)>1 ORDER BY occurrences DESC;
```



order_id	occurrences

```sql
-- 05 REPEATED CUSTOMER UNIQUE ID (may be legitimate)
SELECT customer_unique_id, COUNT(*) occurrences FROM customers
GROUP BY customer_unique_id HAVING COUNT(*)>1 ORDER BY occurrences DESC LIMIT 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

customer_unique_id	occurrences
8d50f5eadf50201ccdcedfb9e2ac8455	17
3e43e6105506432c953e165fb2acf44c	9
1b6c7548a2a1f9037c1fd3ddfed95f33	7
6469f99c1f9dfae7733b25662e7f1782	7
ca77025e7201e3b30c44b472ff346268	7
47c1a3033b8b77b3ab6e109eb4d5fdf3	6
dc813062e0fc23409cd255f7f53c7074	6
63cfc61cee11cbe306bff5857d00bfe4	6
f0e310a6839dce9de1638e0fe5ab282a	6
de34b16117594161a6a89c50b289d35a	6

```sql
-- 04 DUPLICATE CUSTOMER ID
SELECT customer_id, COUNT(*) occurrences FROM customers
GROUP BY customer_id HAVING COUNT(*)>1 ORDER BY occurrences DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	occurrences

```sql
-- 03 PAYMENTS NULL COMPLETENESS
SELECT COUNT(*) total_rows,
SUM(order_id IS NULL) null_order_id,
SUM(payment_sequential IS NULL) null_payment_sequential,
SUM(payment_type IS NULL OR TRIM(payment_type)='') missing_payment_type,
SUM(payment_installments IS NULL) null_installments,
SUM(payment_value IS NULL) null_payment_value
FROM order_payments;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_rows	null_order_id	null_payment_sequential	missing_payment_type	null_installments	null_payment_value
103886	0	0	0	0	0

```sql
-- 02 ORDERS NULL COMPLETENESS
SELECT COUNT(*) total_rows,
SUM(orde
SUM(customer_id IS NULL) null_customer_id,
SUM(order_status IS NULL OR TRIM(order_status)='') missing_status,
SUM(order_purchase_timestamp IS NULL) null_purchase_timestamp,
SUM(order_approved_at IS NULL) null_approved_at,
SUM(order_delivered_carrier_date IS NULL) null_carrier_date,
SUM(order_delivered_customer_date IS NULL) null_delivered_customer_date,
SUM(order_estimated_delivery_date IS NULL) null_estimated_delivery_date
FROM orders;
```

Execute the statement under the keyboard cursor

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_rows	null_order_id	null_customer_id	missing_status	null_purchase_timestamp	null_approved_at	null_carrier_date	null_delivered_cu
99441	0	0	0	0	0	0	0

```sql
-- 01 CUSTOMERS NULL COMPLETENESS
SELECT
SUM(customer_id IS NULL) null_customer_id,
SUM(customer_unique_id IS NULL) null_customer_unique_id,
SUM(customer_zip_code_prefix IS NULL) null_zip,
SUM(customer_city IS NULL OR TRIM(customer_city)='') missing_city,
SUM(customer_state IS NULL OR TRIM(customer_state)='') missing_state
FROM customers;
```

Execute the statement under the keyboard cursor

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_rows	null_customer_id	null_customer_unique_id	null_zip	missing_city	missing_state
99441	0	0	0	0	0

```sql
-- 00 ROW COUNTS
SELECT 'customers' table_name, COUNT(*) total_records FROM customers
UNION ALL SELECT 'orders', COUNT(*) FROM orders
UNION ALL SELECT 'order_payments', COUNT(*) FROM order_payments;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

table_name	total_records
customers	99441
orders	99441
order_payments	103886